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Advanced Sensitivity Analysis for Pipeline Free-Span: Leveraging Python Framework with DNV Compliance

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Abstract

Subsea pipelines are critical components of offshore oil and gas infrastructure, often traversing uneven seabed terrains that result in unsupported spans. These free spans are susceptible to vortex-induced vibrations (VIV), which can lead to fatigue damage, local buckling, and ultimately, structural failure if not properly assessed and mitigated. Accurate prediction of allowable free span lengths is therefore essential to ensure the long-term integrity and operational safety [1, 2].

The DNVGL-RP-F105 guideline remains the industry standard for assessing free-spanning pipelines under VIV. However, its application involves numerous interdependent parameters—ranging from hydrodynamic loading and seabed stiffness to damping ratios and current profiles—making the design process both complex and sensitive to input variability [3, 4]. Recent studies have emphasized the need for robust sensitivity analysis frameworks to identify dominant parameters and reduce conservatism in design [5, 6].

Global sensitivity analysis (GSA), particularly using Sobol indices, has emerged as a powerful tool for quantifying the influence of input variables on model outputs [7, 8]. Saltelli's sampling method, a variance-based approach, enables efficient exploration of high-dimensional input spaces and has been successfully applied in structural and geotechnical engineering contexts [9, 10]. In offshore pipeline design, GSA facilitates rapid scenario testing and risk quantification, enabling engineers to prioritize design variables and optimize resource allocation [11, 12].

Introduction

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Several researchers have applied GSA to offshore systems. For instance, Yang and Li [13] used Sobol indices to assess the impact of seabed stiffness and damping on fatigue life predictions. Similarly, Gao et al. [14] demonstrated the effectiveness of nonlinear soil modeling in evaluating VIV response. These studies underscore the importance of integrating probabilistic methods into deterministic design frameworks.

Hydrodynamic loading, particularly under extreme current conditions, has been shown to significantly influence VIV response [15, 16]. Modal damping ratios, both structural and soil-induced, also play a critical role in controlling vibration amplitudes and fatigue accumulation [17, 18]. Moreover, the interaction between pipeline and seabed—governed by soil stiffness, gap ratio, and embedment depth—introduces nonlinearities that are often overlooked in simplified models [19, 20].

In response to these challenges, this study presents a novel Python-based computational framework that integrates global sensitivity analysis with the DNVGL-RP-F105 methodology. The framework uses a convergence criterion based on the pipeline's fundamental frequency, and stringent tolerance (10^{-6}) ensuring that the dynamic response is captured with high fidelity. This approach mitigates the risk of unconservative predictions and downstream error propagation, as highlighted by recent works on frequency-domain fatigue analysis [21, 22] and provides a transparent and automated means of identifying the most influential design parameters.

The novelty of this work lies in its automation and transparency. By embedding the entire analysis pipeline—from input sampling to sensitivity computation—into a single script, the framework significantly reduces turnaround time and enhances reproducibility. This is particularly valuable for offshore engineering teams operating under tight project schedules and regulatory constraints.

The remainder of this manuscript is structured as follows: Section 2 details the mathematical background and the DNVGL-RP-F105 compliance strategy. Section 3 outlines the development of the Python framework and the implementation of the sensitivity analysis. Section 4 presents the results of the case studies, highlighting key observations and engineering implications. Finally, Section 5 discusses the broader impact of the findings and outlines potential avenues for future research and development.

VIV Screening: DNV-Complaint Mathematical Framework

Objective

To determine the maximum allowable free span length for a subsea pipeline such that vortex-induced vibrations (VIV) do not occur during the design life. This is achieved by comparing the natural frequency of the span with the onset reduced velocity criteria for both in-line and crossflow directions. The input parameters, calculation flow and important mathematical description is provided in the following subsections.

Input Parameters

The design input parameters involved in the analysis are listed below

Geometric and Material Properties

D : Outer diameter of the pipe (including coatings and marine growth)

t_s : Steel wall thickness

E : Young's modulus of steel

ν : Poisson's ratio of steel

ρ_s : Steel density

ρ_w : Seawater density

ρ_c : Concrete density

ρ_m : Marine growth density

α_T : Thermal expansion coefficient

CSF : Concrete Stiffening Factor

Damping Ratios

ζ_{str} : Structural damping

ζ_{hyd} : Hydrodynamic damping

ζ_{soil} : Soil damping

ζ_{total} : Total damping = $\zeta_{str} + \zeta_{soil} + \zeta_{hyd}$

Environmental Conditions

U_c : Current velocity at pipe level

U_w : Wave-induced velocity

H_s : Significant wave height

T_p : Peak wave period

θ : Flow direction relative to pipeline axis

z_0 : Seabed roughness

Soil Properties

K_L : Lateral soil stiffness

K_V : Vertical soil stiffness

ζ_{soil} : Soil damping ratio

Safety Factors

γ_{on}^{IL} : Onset factor for in-line VIV

γ_{on}^{CF} : Onset factor for crossflow VIV

γ_k : Stability parameter safety factor

Added Mass

The added mass m_a represents the inertia of the surrounding fluid that moves with the pipeline:

$$m_a = C_a \rho_w \frac{\pi D^2}{4} \quad (1)$$

where C_a is the added mass coefficient, ρ_w is the water density, and D is the outer diameter of the pipeline.

Hydrodynamic Loading

Hydrodynamic forces acting on the pipeline are modeled using the Morison equation, which accounts for both drag and inertia components and considering pipe velocity:

$$F(t) = \frac{1}{2} \rho_w C_D D \left(U(t) - \frac{dy(t)}{dt} \right) \left| U(t) - \frac{dy(t)}{dt} \right| + \rho_w C_M \frac{\pi D^2}{4} - m_a \frac{dy(t)}{dt} \quad (2)$$

where:

ρ_w is the water density,

C_D is the drag coefficient,

C_M is the inertia coefficient,

D is the pipeline diameter,

y is the pipe lateral displacement

$U(t)$ is the total flow velocity at time t .

The total velocity $U(t)$ includes contributions from both wave-induced and current-induced components. The wave particle velocity is computed using Airy wave theory:

$$U_w(z) = \frac{\pi H_s}{T_p} \frac{\cosh(k(z+h))}{\sinh(kh)} \quad (3)$$

where:

H_s is the significant wave height,

T_p is the peak wave period,

$k=(2\pi)/(L)$ is the wave number,

L is the wavelength,

h is the water depth,

z is the vertical position relative to the still water level.

The current velocity profile is modeled according to DNV RP-C2025 as a power-law distribution:

$$U_c(z) = U_{ref} \left(\frac{z}{z_{ref}} \right)^\alpha \quad (4)$$

where:

U_{ref} is the reference current velocity at height z_{ref} ,

α is the velocity profile exponent (typically 1/7 for turbulent boundary layers).

Axial Tension

The effective axial force S_{eff} accounts for internal and external pressures, thermal expansion, and axial tension:

$$S_{eff} = T - p_i A_i + p_e A_e - E \alpha_T \Delta T \quad (5)$$

where:

T : Axial tension

p_i, p_e : Internal and external pressures

A_i, A_e : Internal and external cross-sectional areas

E : Young's modulus

α_T : Thermal expansion coefficient

ΔT : Temperature difference

Reduced Velocity

The reduced velocity V_R is a non-dimensional parameter that characterizes the flow-induced vibration regime:

$$V_R = \frac{U_c + U_w}{f_n D} \quad (6)$$

where U_c and U_w are the current and significant wave-induced velocities, f_n is the natural still water frequency, and D is the pipeline diameter.

The current flow velocity ratio is defined as

$$\alpha = \frac{U_c}{U_c + U_w} \quad (7)$$

with increase of this number flow regime translates to a pure current condition.

Natural Frequency

The natural frequency f_n of the pipeline is influenced by stiffness, mass, and axial force:

$$f_n = C_1 \sqrt{1 + CSF} \cdot \sqrt{\frac{EI}{m_e L_{eff}^4} \left(1 + \frac{S_{eff}}{P_{cr}} + C_3 \left(\frac{\delta}{D} \right)^2 \right)} \quad (8)$$

where:

C_1, C_3 : Boundary condition coefficient

CSF : Concrete stiffness factor

EI : Flexural rigidity

m_e : Effective mass per unit length

P_{cr} : Critical buckling load

δ : Static deflection, significant for crossflow vibration. Can be calculated as per 6.8.6 DNVGL-RP-F105, if not given.

Static Effective Length

The effective length L_{eff} is a function of the actual span length L and a coefficient C_{eff} that depends on boundary conditions and can be found at Table 6–1, 6.8.7 DNVGL-RP-F105:

$$L_{eff} = C_{eff} \cdot L \quad (9)$$

The L_{eff}/L term is dependent on β which is a function of vertical/horizontal, static/dynamic soil stiffness. Refer 6.8.8 DNVGL-RP-F105.

Onset Criteria

The onset of VIV is determined by comparing the fundamental frequency/reduced velocity to threshold values:

In-Line Direction.

$$\frac{f_{IL}}{\gamma_{IL}} = \frac{U_{C,100yr}}{V_{R,on}^{IL} D} \left(1 - \frac{L}{250} \right)^{\frac{1}{\alpha}} \quad (10)$$

where, $V_{R,on}^{CF}$ is calculated as detailed in 4.6.4 DNVGL-RP-F105.

Crossflow Direction.

$$\frac{f_{CF}}{\gamma_{CF}} = \frac{U_{C,100yr} + U_{W,1yr}}{V_{R,on}^{CF} D} \quad (11)$$

$$V_{R,on}^{CF} = \frac{3 \cdot \psi_{proxi} \cdot \psi_{trench}}{\gamma_{on}^{CF}} \quad (12)$$

where:

$\psi_{proxi} = \frac{1}{5} (4 + 1.25 \cdot \text{gap ratio})$ for gap ratio < 0.8, 1 for other conditions, and $\psi_{trench} = 1 + 0.5 \cdot \frac{\text{trench depth}}{D}$ are correction factors for onset crossflow due to seabed proximity and trench effect, respectively.

$\gamma_{on}^{IL}/\gamma_{on}^{CF}$ are safety factors on onset value for in-line/crossflow reduced velocity.

Stability Parameter

The stability parameter K_S quantifies the damping capacity relative to hydrodynamic excitation:

$$K_S = \frac{4\pi m_e \zeta_{total}}{\rho_W D^2} \quad (13)$$

where ζ_{total} is the total damping ratio.

Allowable Span Length

The allowable span length is determined iteratively to ensure that the fundamental frequency/reduced velocity remains below the onset threshold for both in-line and crossflow directions. This criterion ensures that the pipeline remains outside the VIV excitation regime, thereby mitigating fatigue damage and ensuring structural integrity.

Python Framework and Sensitivity Analysis

The computational backbone of this study is a robust Python-based framework meticulously engineered to automate the entire sensitivity analysis workflow. This framework is tailored to evaluate the dynamic response of free-spanning subsea pipelines subjected to vortex-induced vibrations (VIV), a phenomenon that can significantly impact the structural integrity and fatigue life of offshore infrastructure. By integrating probabilistic variability in key design parameters—such as span length, seabed stiffness, current velocity, and pipeline diameter—the framework ensures a comprehensive exploration of uncertainty in pipeline behavior.

Crucially, the methodology adheres to the guidelines set forth in **DNVGL-RP-F105**, a widely recognized standard for assessing free spans in subsea pipelines. The framework leverages Saltelli Sampling and global sensitivity analysis techniques, including Sobol indices, to quantify the influence of each parameter on the pipeline's response. This enables engineers to identify critical factors affecting fatigue life and optimize design strategies accordingly. The automation not only enhances computational efficiency but also ensures repeatability and traceability in engineering assessments.

Global Sensitivity Analysis via Hoeffding-Sobol Decomposition

At the core of the sensitivity analysis lies the Hoeffding-Sobol decomposition, a variance-based technique that decomposes the total output variance into contributions from individual input variables and their interactions. Let the model output be denoted by $Y=f(\mathbf{X})$, where $\mathbf{X}=(X_1, X_2, \dots, X_k)$ is the vector of k independent input variables. The total variance $\text{Var}(Y)$ can be decomposed as:

$$\text{Var}(Y) = \sum_{i=1}^k V_i + \sum_{i < j} V_{ij} + \sum_{i < j < l} V_{ijl} + \dots + V_{1,2,\dots,k} \quad (14)$$

where $V_i = \text{Var}_{X_i} [E(Y|X_i)]$ is the first-order effect of variable X_i , and V_{ij} , V_{ijl} , etc., represent higher-order interaction effects.

The corresponding Sobol sensitivity indices are defined as:

$$S_i = \frac{V_i}{\text{Var}(Y)}, \quad S_{T_i} = 1 - \frac{\text{Var}_{\sim X_i} [E(Y | \sim X_i)]}{\text{Var}(Y)} \quad (15)$$

where S_i is the first-order Sobol index and S_{T_i} is the total-order index, capturing all effects involving X_i .

The second-order Sobol index S_{ij} quantifies the interaction effect between two input variables X_i and X_j on the output variance, excluding their individual contributions.

$$S_{ij} = \frac{V_{ij}}{\text{Var}(Y)}, \quad S_{T_{ij}} = 1 - \frac{\text{Var}_{X_i, X_j} [E(Y | X_i, X_j)] - V_i - V_j}{\text{Var}(Y)} \quad (16)$$

where: V_{ij} : The variance contribution due to the interaction between X_i and X_j .

$\text{Var}(Y)$: Total variance of the model output.

$E(Y|X_i, X_j)$: Expected value of the output conditioned on both X_i and X_j .

V_i , V_j : First-order variance contributions of X_i and X_j , respectively.

Saltelli Sampling for Efficient Variance Estimation

Saltelli's sampling scheme is a widely adopted method for computing Sobol sensitivity indices, which quantify the contribution of each input variable to the output variance of a model. This scheme is particularly effective because it leverages quasi-random low-discrepancy Sobol sequences, which are designed to uniformly cover the input space with fewer samples compared to purely random sampling. This uniformity enhances the convergence rate of the Sobol index estimators, making the sensitivity analysis more accurate and efficient.

For a model with k input parameters and N base samples, Saltelli's method requires a total of $N(2k+2)$ model evaluations. This formula arises from the need to compute both first-order and total-order indices, which involve evaluating the model under various combinations of fixed and varied inputs. Despite the seemingly large number of evaluations, this approach ensures a comprehensive exploration of the input space, capturing both individual and interaction effects among inputs.

Moreover, Saltelli's scheme is compatible with parallel computing, allowing for further acceleration of the sensitivity analysis process. Its robustness and efficiency make it a preferred choice in engineering, environmental modeling, and other domains where understanding input-output relationships is critical.

Integration with DNVGL-RP-F105 Methodology

The framework integrates the DNVGL-RP-F105 compliant fatigue assessment by embedding the following key components:

- Hydrodynamic force coefficients C_L and C_D as functions of reduced velocity V_R .
- Modal damping ratios ζ_m , including structural and soil-induced components.
- Soil-pipe interaction parameters such as gap ratio G/D , soil stiffness k_s , and embedment depth

The dynamic response is computed iteratively until convergence is achieved on the fundamental natural frequency f_1 , with a tolerance of 10^{-6} , ensuring numerical stability and physical fidelity.

Framework Architecture and Automation

The Python framework is modular and fully automated, consisting of the following stages:

- Input Sampling: Generation of input matrices using Saltelli's method via the SALib library.
- Model Evaluation: Execution of the DNV-compliant VIV response model for each sample described in Section -2.
- Variance Decomposition: Computation of Sobol indices using Monte Carlo estimators.
- Visualization and Reporting: Automated generation of sensitivity plots and ranked parameter influence.

Engineering Implications

The framework enables engineers to isolate key drivers of variability—such as current velocity profiles, modal damping ratios, and soil-pipe interaction characteristics—thereby facilitating:

- Rapid scenario testing under uncertain environmental and geotechnical conditions
- Quantitative risk assessment and prioritization of design parameters
- Reduction of conservatism in design without compromising safety

This integration of global sensitivity analysis with a DNV-compliant fatigue assessment model represents a significant advancement in offshore pipeline design methodology, offering both transparency and computational efficiency.

Results & Discussion - Implementation on Real Case Study

This section details the results of the presently developed framework. First, the results are validated with the existing real project data. Subsequently, a convergence study is performed in order to decide the number of samples required for the present scenario testing, optimizing between calculation accuracy and computation cost. Finally, global sensitivity analysis is performed considering the multiple design input parameters and the sensitivity indices are observed and discussed.

Benchmarking Study

A benchmarking study is performed in order to validate the developed free-span screening Python framework with help of real project data. The input parameters and the results are presented in [Table 1](#) and [Table 2](#), respectively.

Table 1—Input data for benchmarking study for the developed Python framework with real project data.

Input Design Parameters			
Pipeline Data			
Nominal outer steel diameter (mm)	457.2	Wall thickness (mm)	22.2
Marine growth thickness (mm)	0	Marine growth density (kg/m ³)	1019.36
Internal layer thickness (mm)	0	Internal layer density (kg/m ³)	8140
Concrete coating thickness (mm)	25	Concrete coating density (kg/m ³)	2700
Concrete water absorption (%)	0	Concrete slippage factor	0.4
Direction of pipeline (deg.)	0	Content density for pipe in operation (kg/m ³)	176.1
Material resistance factor	1.15	Fabrication factor	0.85
Environmental Load			
Wave direction (deg.)	90	Current direction (deg.)	90
Significant wave ht. 1-yr (m)	1.3	Current velocity, 1-yr (m/s)	0.35
Significant wave ht. 10-yr (m)	1.4	Current velocity, 10-yr (m/s)	0.41
Significant wave ht. 100-yr (m)	1.6	Current velocity, 100-yr (m/s)	0.46
Peak time period, 1-yr (s)	5.5	Reference ht. for current (m)	1
Peak time period, 10-yr (s)	5.9	Water depth (m)	2
Peak time period, 100-yr (s)	6.1	Seawater density (kg/m ³)	1025
Max wave ht. 1-yr (m)	1.6	Maximum time period, 1-yr (s)	5.1
Max wave ht. 10-yr (m)	1.6	Maximum time period, 10-yr (s)	5.5
Max wave ht. 100-yr (m)	1.6	Maximum time period, 100-yr (s)	5.7
Input Stresses			
Effective axial force, as-laid/empty (kN)	0	Route bend stress (MPa)	0
Effective axial force, hydrotest (kN)	-1358	Derating SMYS (MPa)	0
Effective axial force, operating (kN)	-3485	Derating SMTS (MPa)	0
Operating Data			
Pressure for hydrotest condition (bar)	265.65	Temperature for as-laid condition (°C)	17.9
Pressure for operating condition (bar)	230.00	Temperature for hydrotest condition (°C)	34.5
		Temperature for operation condition (°C)	36
Additional Conditions			
Span gap ratio	1.94	Boundary condition	Single span
Safety class (empty/hydrotest)	low	Safety class (operating)	high
Structural damping ratio	0.02	Hydrodynamic damping ratio	0
Peak enhancement factor	3.65		

Table 2—Comparison of the final output free-span length: developed Python framework with real project data.

Maximum allowable span	Condition	Project Data	Python Framework	Error (%)
In-line (m)	Empty (EMP)	21.863	21.863	0
	Hydrotest (HYD)	21.118	21.118	0
	Operation (OPE)	16.719	16.719	0
Crossflow (m)	Empty (EMP)	28.384	28.384	0
	Hydrotest (HYD)	25.306	25.306	0
	Operation (OPE)	21.638	21.638	0

Global Sensitivity Analysis

Once the Python framework is validated with real project data, next some input parameters are identified for subsequent sensitivity analysis. These input parameters are: 1. Current velocity (1 yr), 2. Current velocity (10 yr), 3. Current velocity (100 yr), 4. Span gap ratio, and 5. Structural damping ratio.

A convergence study is performed prior to the global sensitivity analysis, and no significant change in sensitivity index data is observed between 2^{10} and 2^{11} sample points. Therefore, in subsequent analysis for each condition, allowable in-line and crossflow free span lengths are estimated at 1024 (2^{10}) sample points between the upper and lower bound values of the input parameters tabulated below.

Table 3—Lower and upper bound values for the sampling parameters.

Parameter	Lower Bound	Upper Bound
Current velocity (1 yr), m/s	0.01	1.0
Current velocity (10 yr), m/s	0.01	1.5
Current velocity (100 yr), m/s	0.01	2.0
Span gap ratio	1	5.0
Structural damping ratio	0.002	0.05

Discussion

The global sensitivity analysis results, illustrated in [Figures 1](#) and [2](#), offer valuable insights into the relative influence of input parameters across three operational states—Empty (EMP), Hydrotest (HYD), and Operational (OPE). The analysis was performed separately for inline and crossflow vortex-induced vibrations (VIV), using a sample size of 1024 and computing first-order (S_1), second-order (S_2), and total-order (S_T) Sobol indices.

- Current velocity with 100 years return-period exhibits the highest total-order index S_T , indicating it is the most influential parameter in governing in-line and crossflow VIV response. However, the extreme and rare occurrence of this event should also be considered. Its moderate S_1 value indicates moderate direct effect, while a greater S_2 suggests strong interaction with other parameters.
- Span Gap Ratio and Structural Damping Ratio exhibit significant direct sensitivity in both in-line and crossflow VIV analyses with S_1 values close to 0.5. Span Gap Ratio demonstrates significant secondary interaction with S_2 value over 0.8, suggesting that Span Gap Ratio interacts with other parameters in shaping the vibrational behavior. These results emphasize that this ratio along with environmental load is a key contributor to in-line and crossflow VIV control and must be carefully characterized in design and analysis. However, coupled ($S_2 \sim 0.3$) and total contribution ($S_T \sim 0.2$) for Structural Damping Ratio is found to be lesser.
- Current Velocity Profiles (1-year, 10-year, and 100-year return periods) show increasing total contributions to output variance. The influence S_T increases with return period, reflecting the

growing importance of extreme environmental loading in overall free-span estimation. Though the first-order effects are quite less for 1-year and 10-year current velocities, their second-order interactions are significant, as indicated by their respective S_1 and S_2 values.

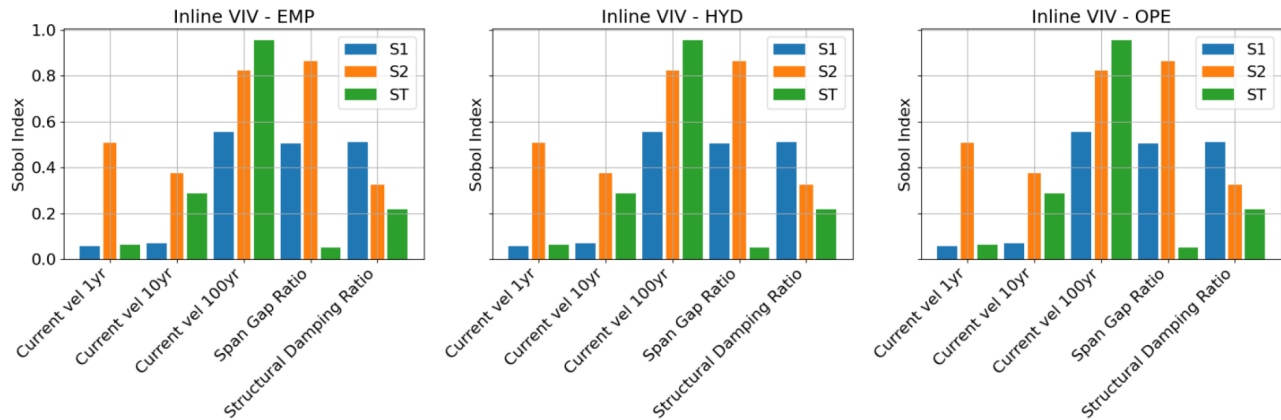


Figure 1—First order (S_1), second-order (S_2) and total Sobol sensitivity (S_T) index of the input parameters with sample size 1024 for in-line VIV: (a)EMP: empty or as-laid condition; (b)HYD: hydrotest condition; (c)OPE: operational condition.

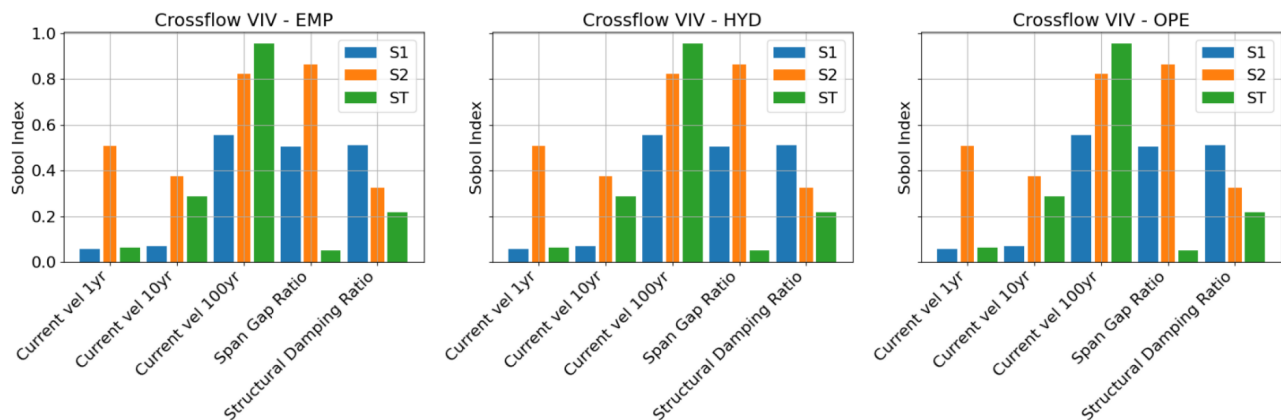


Figure 2—First order (S_1), second-order (S_2) and total Sobol sensitivity (S_T) index of the input parameters with sample size 1024 for crossflow VIV: (a) EMP: empty or as-laid condition; (b) HYD: hydrotest condition; (c) OPE: operational condition.

This sensitivity landscape simplifies the interpretation and prioritization of design parameters. It highlights the dominant role of geometric and environmental factors—particularly the span gap ratio and long-term current profiles—in shaping crossflow VIV behavior, while damping effects remain secondary.

Engineering Implications

These findings have direct implications for offshore pipeline design:

- **Design Prioritization:** Parameters with high S_1 and S_T values should be prioritized for accurate measurement and modeling.
- **Scenario Testing:** The framework enables rapid identification of critical conditions where parameter interactions may lead to conservative or unconservative predictions.
- **Risk Quantification:** By quantifying the contribution of each parameter to output uncertainty, the framework supports probabilistic risk assessments aligned with DNVGL-RP-F105.

Conclusions and Future Work

Conclusions

This study presents a robust and automated Python-based framework for global sensitivity analysis of free-spanning subsea pipelines subjected to vortex-induced vibrations. The integration of the DNVGL-RP-F105 fatigue assessment methodology with Sobol sensitivity analysis—using Saltelli's sampling—enables a transparent and computationally efficient approach to uncertainty quantification.

Key contributions include:

- Development of a modular Python framework that embeds input sampling, model evaluation, and sensitivity computation.
- Application of Hoeffding-Sobol decomposition to decompose output variance into first-order, second-order, and total effects.
- Identification of dominant parameters under different operational states for both in-line and crossflow VIV.
- Demonstration of the importance of soil-pipe interaction and hydrodynamic loading in fatigue-sensitive design.

Future Work

Future extensions of this work may include:

- Incorporation of Bayesian Updating: To refine parameter distributions based on field measurements or monitoring data.
- Extension to Multi-span Systems: Capturing modal interactions and span-to-span coupling effects.
- Integration with Machine Learning: Surrogate modeling techniques (e.g., Gaussian Processes or Neural Networks) to reduce computational cost for large-scale Monte Carlo simulations.
- Regulatory Integration: Embedding the framework into digital twin platforms for real-time compliance monitoring with DNV and ISO standards.

The proposed framework thus serves as a foundational tool for advancing reliability-based design and digital engineering practices in offshore pipeline systems.

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